

CHAPTER 10.3

AREA FOR POLAR CURVES

In this Chapter, We will show how to find areas of regions that are bounded by polar curves.

AREA IN POLAR COORDINATES

10.3.3 Area Problem in Polar Coordinates

Suppose that α and β are angles that satisfy the condition $\alpha < \beta \leq \alpha + 2\pi$ and suppose that $f(\theta)$ is continuous and nonnegative for $\alpha \leq \theta \leq \beta$. Find the area of the region R enclosed by the polar curve $r = f(\theta)$ and the rays $\theta = \alpha$ and $\theta = \beta$ (**Figure 10.3.7**).

10.3.4 Area in Polar Coordinates

If α and β are angles that satisfy the condition $\alpha < \beta \leq \alpha + 2\pi$ and if $f(\theta)$ is continuous and either nonnegative or nonpositive for $\alpha \leq \theta \leq \beta$, then the area A of the region R enclosed by the polar curve $r = f(\theta)$ ($\alpha \leq \theta \leq \beta$) and the lines $\theta = \alpha$ and $\theta = \beta$ is

$$A = \int_{\alpha}^{\beta} \frac{1}{2} [f(\theta)]^2 d\theta = \int_{\alpha}^{\beta} \frac{1}{2} r^2 d\theta$$

The hardest part of applying (6) is determining the limits of integration. This can be done as follows:

Area in Polar Coordinates: Limits of Integration

Step 1. Sketch the region R whose area is to be determined.

Step 2. Draw an arbitrary “radial line” from the pole to the boundary curve $r = f(\theta)$.

Step 3. Ask, “Over what interval of values must θ vary in order for the radial line to sweep out the region R ?”

Step 4. Your answer in Step 3 will determine the lower and upper limits of integration.

Example 6

Find the area of the region *in the first quadrant* that is within the cardioid $r = 1 - \cos \theta$.

Example 7

Find the *entire area* within the cardioid of $r = 1 - \cos \theta$.

Example 8

Find the area of the region enclosed by the rose curve $r = \cos 2\theta$.

Example 9

Find the area of the region that is inside of the cardioid $r = 4 + 4\cos \theta$ and outside of the circle $r = 6$.

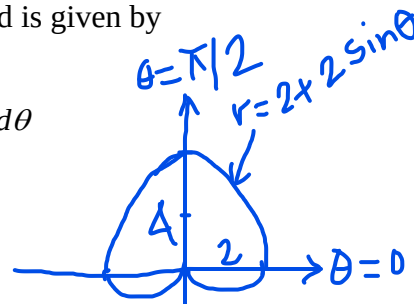
Exercise Set 10.3

29. Find the area of the region that is enclosed by the cardioid $r = 2 + 2 \sin \theta$.

Solution: The area of the region that is enclosed by the cardioid is given by

$$A = \int_0^{2\pi} \frac{1}{2} (2 + 2 \sin \theta)^2 d\theta = 2 \int_0^{2\pi} [1 + 2 \sin \theta + \sin^2 \theta] d\theta$$

$$= 2 \left[\theta - 2 \cos \theta \right]_0^{2\pi} + \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^{2\pi} = 6\pi.$$

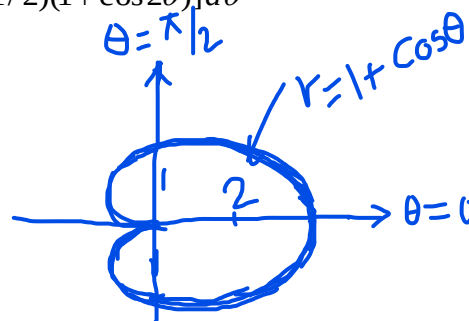


30. Find the area of the region in the first quadrant within the cardioid $r = 1 + \cos \theta$.

Solution: The area of the region in the first quadrant is given by

$$A = \int_0^{\pi/2} \frac{1}{2} (1 + \cos \theta)^2 d\theta = \frac{1}{2} \int_0^{\pi/2} [1 + 2 \cos \theta + (1/2)(1 + \cos 2\theta)] d\theta$$

$$= \frac{1}{2} \left[\frac{3}{2} \theta + 2 \sin \theta + \frac{1}{4} \sin 2\theta \right]_0^{\pi/2} = \frac{3\pi}{8} + 1.$$

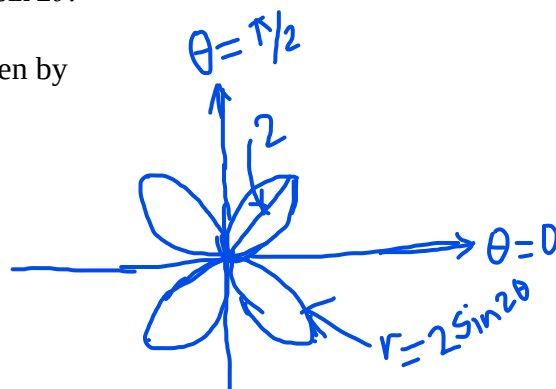


32. Find the area of the region enclosed by the rose $r = 2\sin 2\theta$.

Solution: The area of the petal in the first quadrant is given by

$$A = \int_0^{\pi/2} \frac{1}{2} (2\sin 2\theta)^2 d\theta = \int_0^{\pi/2} [1 - \cos 4\theta] d\theta$$

$$= \left[\theta - \frac{1}{4} \sin 4\theta \right]_0^{\pi/2} = \frac{\pi}{2}.$$



Hence the entire area will be 2π .

40. Find the area of the region outside the cardioid $r = 2 - 2\cos\theta$ and inside the circle $r = 4$.

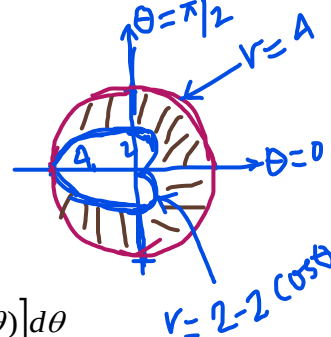
Solution: When the two graphs intersect then

$$2 - 2\cos\theta = 4 \Rightarrow 2\cos\theta = -2 \Rightarrow \cos\theta = -1 \Rightarrow \theta = \pi$$

The desired area is

$$A = 2 \int_0^{\pi} \frac{1}{2} [16 - (2 - 2\cos\theta)^2] d\theta = \int_0^{\pi} [16 - (4 - 8\cos\theta + 4\cos^2\theta)] d\theta$$

$$= 12 \left[\theta + 8\sin\theta \right]_0^{\pi} - 2 \left[\theta + \frac{1}{2} \sin 2\theta \right]_0^{\pi} = 10\pi.$$



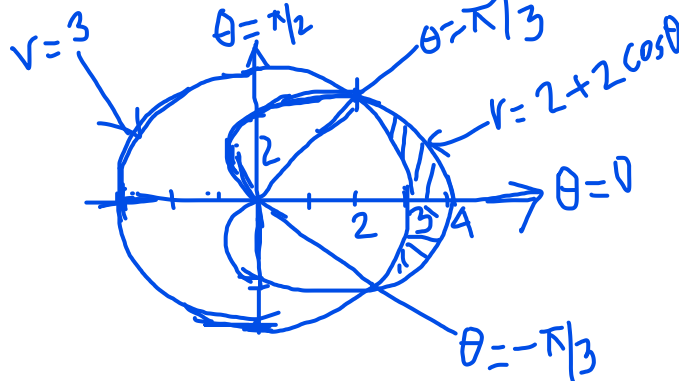
41. Find the area of the region that is inside the cardioid $r = 2 + 2\cos\theta$ and outside the circle $r = 3$.

Solution: When the curves intersect then

$$2 + 2\cos\theta = 3 \Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \theta = \pi/3$$

The desired area is

$$A = 2 \int_0^{\pi/3} \frac{1}{2} [(2 + 2\cos\theta)^2 - 9] d\theta = \int_0^{\pi/3} [8\cos\theta + 4\cos^2\theta - 5] d\theta$$



$$\begin{aligned}
&= [8 \sin \theta - 5\theta]_0^{\pi/3} + 2 \left[\theta + \frac{1}{2} \sin 2\theta \right]_0^{\pi/3} \\
&= \left(8 \frac{\sqrt{3}}{2} - 5 \frac{\pi}{3} \right) + 2 \frac{\pi}{3} + \frac{\sqrt{3}}{2} = 9 \frac{\sqrt{3}}{2} - \pi.
\end{aligned}$$